ECON0019: QUANTITATIVE ECONOMICS AND ECONOMETRICS
EMPIRICAL PROJECT 2025

Instructions

The mark for the empirical project is worth 20% of your total mark for the module.

Please follow the instructions provided below so that we can ensure anonymity in marking and ensure compliance with UCL assessment policies. We will only be able to give you credit for your project if you follow the instructions. If they are not followed, you will receive a mark of zero.

1. Please elect **one group member to submit** the project for the group.
2. All answers must be uploaded via Moodle by 12pm on April 3, 2025.
3. All marking is anonymised. **Do not put your name or student number or group name** anywhere on your submitted answer — either in the document or in the file name.
4. You should submit **one PDF or Word document** that includes: you **answers and explanations** in the main text (including tables and figures, if any), as well as **an appendix with your code** producing the results. If you use software other than Stata, you should state which programme was used. You may optionally include raw statistical output (e.g. Stata log-file) after the code but such output does not substitute for your answers and explanations.
5. Your answers should be no more than 800 words, including footnotes but excluding tables, figures, and the code appendix. You must **state the number of words** at the top of the first page of your submission.
6. Submissions will be checked for plagiarism. By submitting this assessment, you pledge your honour that you have not violated UCL Assessment Regulations which are detailed in the Student Academic Misconduct Procedure, which include (but are not limited to) plagiarism, self-plagiarism, unauthorised collaboration between students, sharing my assessment with another student or third party, access another student's assessment, falsification, contract cheating, and falsification of extenuating circumstances.
7. Please make sure to allow sufficient time should problems arise with submitting your answers via Moodle. Check the submission inbox for confirmation that your answers have been submitted. Once your submission has been accepted you will return to the 'My Submissions' tab where you will be able to see the details of your submission. If your submission is not confirmed for some reason, or you are having issues uploading the document, get in touch with ISD (`servicedesk@ucl.ac.uk`) as soon as possible to figure out what the problem might be.
8. According to UCL's AI guidelines, this assessment falls under Category 1: AI tools cannot be used. The use of AI tools that exceeds that permitted in the assessment brief constitutes Academic Misconduct.

You will be awarded a mark of 0% or Grade F if you (1) do not attempt the summative assessment component or (2) attempt so little of the summative assessment component that it cannot be assessed. Please check the UCL Academic Manual (Chapter 4, Section 3.10) for information on the consequences of not submitting or engaging with any of your assessment components.

If you have extenuating circumstances that affect your ability to engage with any of the module assessment components, please apply for alternative arrangements to the Economics Department as soon as possible. See details in Chapter 2, Section 3 of the Academic Manual and send your request to `economics.ug@ucl.ac.uk`.

If you have a disability or long-term medical condition, you may be entitled to adjustments for assessments. This may include an extension for this essay. Please see Chapter 2, Section 4 of the Academic Manual for information on how to apply for adjustments. Contact the Departmental Tutor, Dr Frank Witte (`f.witte@ucl.ac.uk`) and the UG Admin team (`economics.ug@ucl.ac.uk`). Do not contact the course lecturers about this.

QUESTION:

In "Carbon Taxes and CO_2 Emissions: Sweden as a Case Study" (American Economic Journal: Economic Policy, 2019), Julius Andersson studies how the introduction of a carbon tax and a value-added tax on transport fuel in Sweden in the early 1990s affected emissions of carbon dioxide. As the paper notes, "Sweden was one of the first countries in the world to implement a carbon tax in 1991. It was introduced at the level of US$30 per ton of CO_2 and then successively increased to today's rate of US$132, currently the highest carbon tax in the world. In the year prior, in March of 1990, Sweden also extended the coverage of its existing value-added tax (VAT) to include gasoline and diesel. "

To study this question, the authors compare the evolution of emisions in Sweden before and after the tax introduction to that in a "synthetic Sweden". Intuitively, the emissions for the latter are obtained by a weighted average of OECD countries that emulates a few important variables (e.g., emissions and other variables such as GDP per capita) from Sweden prior to the tax increase. Results indicate a reduction of about 11% or 0.29 metric tons of CO_2 per capita in an average year between 1990 and 2005. This is visually depicted in the figure below:

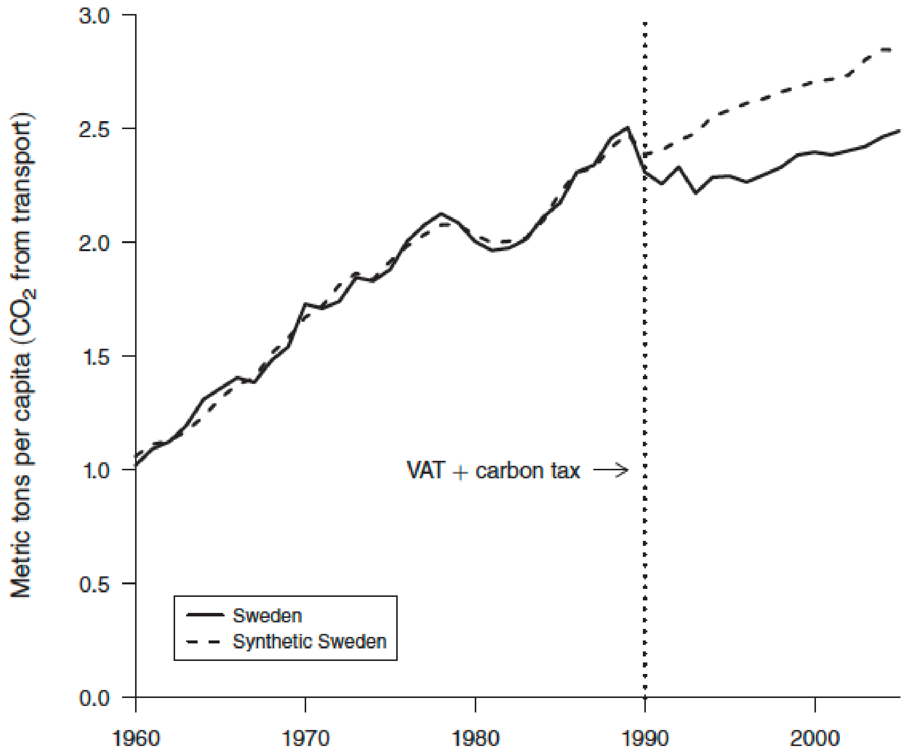

FIGURE 4. PATH PLOT OF PER CAPITA CO_2 EMISSIONS FROM TRANSPORT DURING 1960–2005: SWEDEN VERSUS SYNTHETIC SWEDEN

Aside from the exercise above, the paper also investigates the sensitivity of fuel consumption to the carbon tax and other components of price. When the carbon tax was introduced, it complemented an existing energy tax to the price of petrol and diesel (which had been in place since at least the 1930s). With the inclusion of the VAT in 1990, the retail price of petrol or diesel at time t (p^*_t) consists of a tax-exclusive price (p_t), the energy tax ($\tau_{t,\text{energy}}$), the carbon tax (τ_{t,CO_2}) and the value-added tax (VAT_t):

$$p^*_t = (p_t + \tau_{t,\text{energy}} + \tau_{t,CO_2})\text{VAT}_t. \tag{1}$$

The figure below shows the evolution of the (inflation-adjusted) components through time:

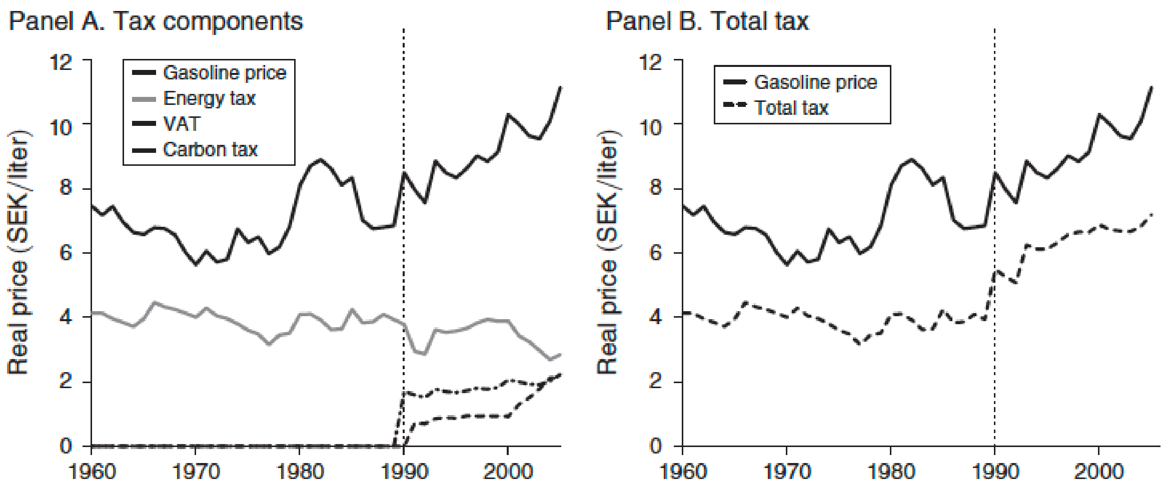

FIGURE 1. GASOLINE PRICE COMPONENTS IN SWEDEN 1960–2005

Petrol demand is then modeled according to the following log-linear specification:

$$\ln(c_t) = \alpha + \beta_1 p_t^v + \beta_2 \tau_{t,CO_2}^v + \text{other factors}, \tag{2}$$

where c_t is the consumption of petrol, $p_t^v = (p_t + \tau_{t,\text{energy}})\text{VAT}_t$ is the carbon tax-exclusive price, and $\tau_{t,CO_2}^v = \tau_{t,CO_2} \times \text{VAT}_t$ is the carbon tax (with VAT). The coefficients β_1 and β_2 thus represent (semi-)elasticities for the consumption of petrol. [1]

The Stata data file `ECON0019_empirical_project_2025.dta` contains annual observations of the variables of interest for the period 1970-2011. Specifically:

- `log_gas_cons` – log-petrol consumption ($\ln(c_t)$)
- `log_gas_cons_cens` – log-petrol consumption ($\ln(c_t)$) censored from below at 5.9 (to be used in Question 9)
- `real_carbontaxexclusive_with_vat` – carbon-tax exclusive gasoline price (p_t^v)
- `real_carbontax_with_vat` – carbon tax (τ_{t,CO_2}^v)
- `real_energytax_with_vat` – energy tax ($\tau_{t,\text{energy}}$)
- `real_oil_price_sek` – (Brent) crude oil price

[1] This is immaterial for this exercise, but because the paper also establishes that taxes appear to be fully passed through to consumers, these price and tax elasticities can be considered as demand elasticities.

- `year` – year

We simplify some of the analysis from the paper.

Some Stata hints:

- You can obtain TSLS estimates either with the command `ivregress` or with the command `ivreg2`.
- Command `test` allows you to compute F-statistics and perform two-sided tests on (single or multiple) coefficients or their linear combinations.
- Before executing time series data commands, you must declare that your data set is a time series using the command `tsset`. For this particular dataset, the command would be `tsset year`. (The variable `year` contains the year of each observation in the Stata data file)
- Command `estat ic` after a regression computes the AIC and BIC for the preceding model.
- Before executing time series commands, you must declare a time variable using `tsset`. The time series operator `L(j/p)varname` includes lags j through p as regressors. The operator `d.varname` computes the first difference. `ac` plots the autocorrelation function.
- Type `help command` to get more details on how a particular command works, e.g. `help test`.

Answer the following questions:

1. Plot each of the three time series $\ln(c_t)$, p_t^v and τ_{t,CO_2}^v as functions of time. Also compute the summary statistics for the data (i.e., average, standard deviation for each of the three time series). Finally, plot the first 10 sample autocorrelations for each of the three time series. Comment on your findings.
2. To accommodate for a potential time trend, first regress each of the three time series $\ln(c_t)$, p_t^v and τ_{t,CO_2}^v against a constant and a (linear) time trend (i.e., the variable `year`) and save the residuals. Next, as an initial examination of the relationship between petrol consumption and (carbon tax-exclusive) prices and the carbon tax, produce a scatterplot of the de-trended (log-) consumption against de-trended (carbon tax-exclusive) price and a scatterplot of de-trended (log-) consumption against the de-trended carbon tax. Do you see any strong relationships between the variables?
3. First, estimate a (simple) regression without a constant of the detrended (log-)consumption (i.e., the residual from the regression above on a constant and a time trend) on the detrended (carbon tax-exclusive) price (i.e., the residual from the regression above on a constant and a time trend).

Next, estimate a regression of (log-)consumtption on (carbon tax-exclusive) price and a linear time trend. Compare and explain the two sets of results. *Here and in the following questions, until instructed otherwise, you should treat the regression errors as having no autocorrelation when computing standard errors and carrying out inference.*

4. To estimate the model in (2), run a regression of (log-)consumption on a constant, (carbon tax-exclusive) prices, the carbon tax and a (linear) time trend. Report and comment on your findings. In particular, explain the difference between the estimated coefficient on p_t^v obtained here and the initial estimate of the same coefficient obtained in Question 3.

5. Test whether $\beta_1 = \beta_2$ using the estimation above.

6. As noted by the author, "There is a risk, however, that the results are biased because of omitted variables, anticipatory effects, or the endogeneity of gasoline prices: that gasoline consumption affects the gasoline price and not just the other way around. Endogeneity is arguably a lesser risk for a small country such as Sweden, compared with larger oil consumers such as the United States, since crude oil prices are set in a global market and changes in demand in Sweden will thus have a negligible impact on the world price. The issue still needs to be addressed, though, since domestic producers (at the retail and refinery level) may adjust their margin, and hence affect the pump price, as a response to local changes in demand." In order to address these concerns, estimate the specification above (i.e., in Question 4) separately first using the energy tax rate as an instrumental variable for the (carbon tax-exclusive) price and second using the (brent) crude oil price as an instrumental variable for the (carbon tax-exclusive) price. What arguments can you provide for the validity (i.e., exogeneity) of each one of these instruments? Report your two sets of results and discuss the relevance of each one of the instrumental variables.

7. Since we have two sets of estimates – based on two potential instruments – we can test their (joint) exogeneity assumption by evaluating how different those estimates are, in a formal test of over-identifying restrictions. Conduct an over-identification test for the regression in (2) (controlling for time trends as in Question 4) where you use both instruments for the (carbon tax-exclusive) price. What do you conclude?
HINT: If you are using the command `ivregress` in Stata, you can run the test using the `estat overid` command post-estimation. If you are on the other hand using the command `ivreg2` in Stata, the overidentification test statistic is displayed at the end of the results table as the Hansen J statistic.

8. How do the TSLS estimates above (in Question 6) compare with the OLS estimates obtained previously (in Question 4)? Under the assumption that the (brent) crude oil price is a valid instrumental variable, test whether the (carbon tax-exclusive) price is exogenous. *HINT: If you are using the command `ivregress` in Stata, you can run the test using the `estat endogenous` command post-estimation. If you are on the other hand using the command `ivreg2` in Stata, you should specify the option `endog(var)` where `var` is the (endogenous) variable you want to*

test for at the end of your command line. (This variable is the carbon tax-exclusive price of gasoline in our case.)

9. Supposed that (log-)consumption is censored below 5.9 (log-)SKK. In other words, though you do observe whether (log-)consumption is below 5.9, you do not observe the exact value for this variable. If this were the case, 8 observations for `log_gas_cons` would fall below the censoring point 5.9 and the variable `log_gas_cons_cens` sets those observations to 5.9.) Using this variable, estimate the model

$$\ln(c_t) = \alpha + \beta_1 p_t^v + \beta_2 \tau_{t,CO_2}^v + \beta_3 t + u_t$$

under censoring and the assumption that the error follows $u_t \sim \mathcal{N}(0, \sigma^2)$. What is the Average Partial Effect (APE) the carbon tax on (log-)consumption? How does it compare with the estimate for β_2 obtained in Question 4?

10. In the previous questions, you were asked to use heteroskedasticity-robust standard errors. However, the included figures show that petrol consumption is serially correlated, and this may be the case even after including a linear time trend. Re-estimate the regression from Question 7. using Newey-West standard errors following the bandwidth rule $m = 0.75T^{\frac{1}{3}}$ (rounding up). How do these standard errors compare to the ones that are robust only to heteroskedasticity? Explain.

11. Did the introduction of the carbon tax change the relationship between prices and petrol consumption in Sweden? Perform a Chow test for the regression

$$\ln c_t = \alpha + \gamma p_t^* + \kappa t + u_t, \tag{3}$$

where p_t^* was defined in equation (1), using the introduction of the tax as the candidate break date and Newey-West standard errors. What do you conclude?